



# 06C THERMOCHEMISTRY

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UNIT 03: WHAT REACTIONS GIVE

CHEMISTRY 1310

# LEARNING OBJECTIVES

## CHAPTER 6

1. Define and apply energy, work, and heat.
2. Describe and apply the concept of conservation of energy.
3. Define and apply state functions and path functions.
4. Apply the concept of state functions to predict the energy changes involved in reactions.
5. Summarize the roles of enthalpy and heat flow in chemical reactions.
6. Calculate pressure-volume work.
7. Calculate the heat required to change the temperature of a substance.
8. Calculate changes in enthalpy and internal energy in chemical reactions.
9. Apply Hess's law to calculate the enthalpy of different reactions.
10. Apply standard enthalpies of formation and Hess's law to calculate the enthalpy of a reaction.

# Energy and Enthalpy

## Chemistry in Context

Recall that heat ( $q$ ) is the transfer of energy *due to a temperature difference* between two objects. Heat will always flow until the two objects reach the same temperature, or a state called thermal equilibrium.

You should recognize that heat is an incredibly easy quantity for us to measure in the lab because it is related to changes in temperature. All we need is a thermometer!

The next several sections are all related to *different ways* to measure or calculate  $\Delta H_{\text{rxn}}$ .

There are three general methods:

- 1) Experimentally through calorimetry (measuring temperature changes)
- 2) Hess's Law: taking advantage of state function properties
- 3) Theoretically through standard enthalpies of formations,  $\Delta H_f^\circ$

# Enthalpy in Chemical Reactions

6.7

## Hess's Law

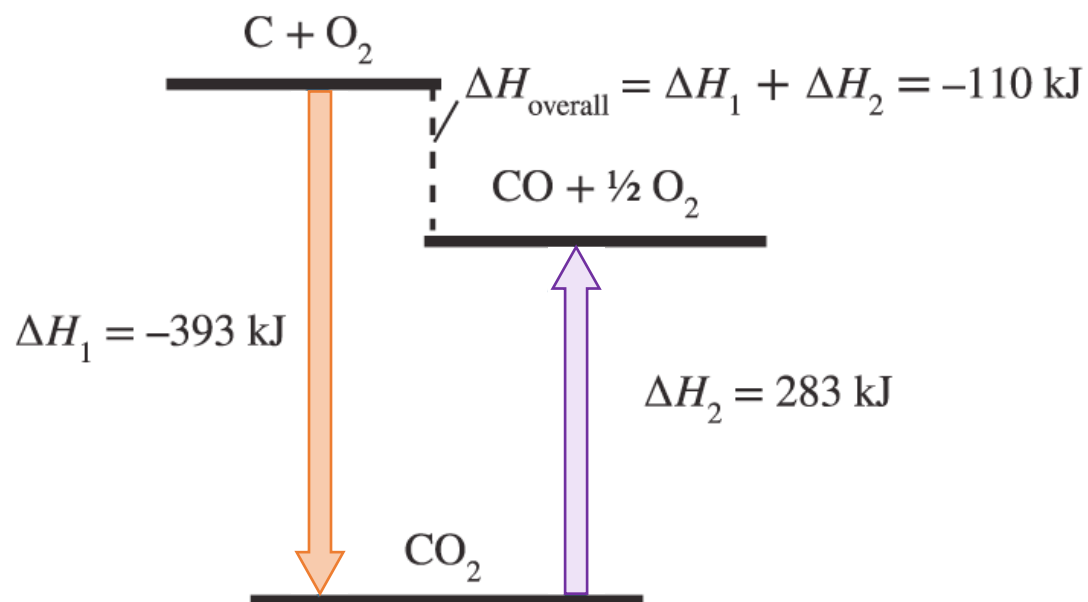
- The enthalpy change for any process is *independent* of the pathway in which the process is carried out. This works because enthalpy is a state function.
- When two or more processes combine to give a resulting process, their enthalpy changes add to give the enthalpy change for the resulting process.

# Enthalpy in Chemical Reactions

6.7

## Hess's Law

- The overall reaction enthalpy is the *sum* of the reaction enthalpies of the steps into which the overall reaction can be divided.



**FIGURE 6.16** Diagram representing the application of Hess's Law for the overall reaction of C to CO.

# Enthalpy in Chemical Reactions

6.7

## General guidelines for applying Hess's Law:

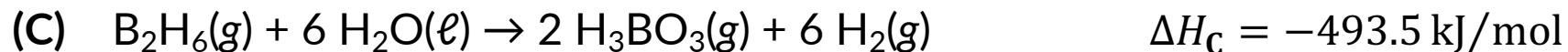
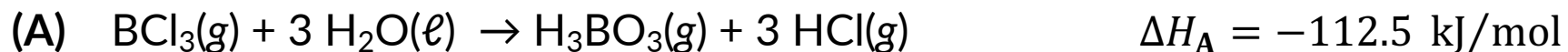
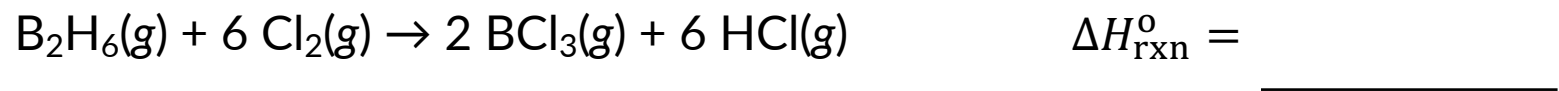
- Whatever you do to the reaction, do the same to its value of  $\Delta H$ .
- If you reverse a reaction, change the sign of  $\Delta H$ .
- If you multiply the coefficients in a reaction by a factor, multiply  $\Delta H$  by the same factor.
- When you add reactions, add their  $\Delta H$ .



## IN-CLASS QUESTION 1

Learning Objective(s): 6.9

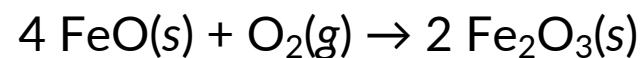
Use reactions **A-C** to determine the standard change in enthalpy ( $\Delta H_{\text{rxn}}^{\circ}$ ) for the following reaction.



## EXAMPLE PROBLEM

Learning Objective(s): 6.9

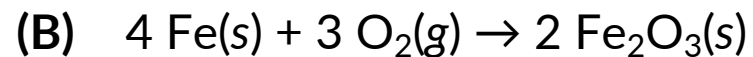
Determine the standard change in enthalpy ( $\Delta H_{\text{rxn}}^{\circ}$ ) for the following reaction.



$$\Delta H_{\text{rxn}}^{\circ} = \underline{\hspace{2cm}}$$



$$\Delta H_{\text{A}}^{\circ} = -544 \text{ kJ/mol}$$



$$\Delta H_{\text{B}} = -1648 \text{ kJ/mol}$$



# Standard Enthalpies of Formation

6.8

## Standard enthalpies/ heats of formation ( $\Delta H_f^\circ$ )

### *a) What is a standard formation reaction?*

- The formation of 1 mole of a substance (in a specified state) from its constituent elements in their standard states.
- Because the definition specifies only 1 mole of product, fractional coefficients are allowed for the reactants (the constituent elements).

### *b) What are standard states of elements?*

- The *most energetically stable* form of the elements under standard conditions.
- Standard conditions are denoted by the  $^\circ$  **symbol** and *defined* as 1 atm for gases and 1 M for aqueous solutions. It also often *implies* 25 °C.
  - Metals → most written as monatomic solids
  - Nonmetals → recall the diatomics  
→ monatomic solids

# Standard Enthalpies of Formation

6.8

## Standard enthalpies/ heats of formation ( $\Delta H_f^\circ$ )

- The enthalpy change associated with the formation of 1 mole of a substance (in a specified phase) from its constituent elements in their standard states.
- Definition implies any element in its standard state must have  $\Delta H_f^\circ = 0$ .
- A table of standard heats of formation for various compounds can be found in Table 6.4 and Appendix A.2 of your textbook.  
*These values will be provided for you when needed.*

# Standard Enthalpies of Formation

6.8

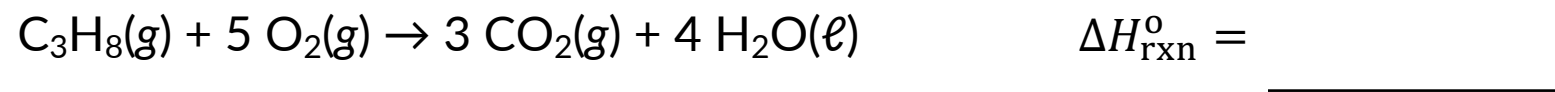
Applying Hess's Law, the standard enthalpies of formation ( $\Delta H_f^\circ$ ) can be used to calculate the standard change in enthalpy for any reaction ( $\Delta H_{\text{rxn}}^\circ$ ).

- This is the *easiest* method to calculate  $\Delta H_{\text{rxn}}^\circ$ !

## EXAMPLE PROBLEM

Learning Objective(s): 6.10

Calculate  $\Delta H_{\text{rxn}}^{\circ}$  for the following reaction using the  $\Delta H_{\text{f}}^{\circ}$  data below.

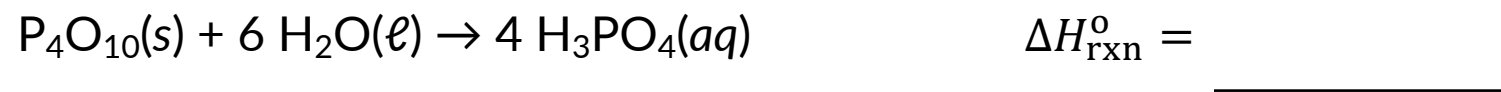


| Substance                        | $\Delta H_{\text{f}}^{\circ} \left( \frac{\text{kJ}}{\text{mol}} \right)$ |
|----------------------------------|---------------------------------------------------------------------------|
| $\text{C}_3\text{H}_8(\text{g})$ | -103.8                                                                    |
| $\text{CO}_2(\text{g})$          | -395.3                                                                    |
| $\text{H}_2\text{O}(\ell)$       | -285.8                                                                    |

## IN-CLASS QUESTION 2

Learning Objective(s): 6.10

Calculate  $\Delta H_{\text{rxn}}^{\circ}$  for the following reaction using the  $\Delta H_{\text{f}}^{\circ}$  data below.



| Substance                           | $\Delta H_{\text{f}}^{\circ} \left( \frac{\text{kJ}}{\text{mol}} \right)$ |
|-------------------------------------|---------------------------------------------------------------------------|
| $\text{P}_4\text{O}_{10}(\text{s})$ | -2984                                                                     |
| $\text{H}_2\text{O}(\ell)$          | -285.8                                                                    |
| $\text{H}_3\text{PO}_4(\text{aq})$  | -1288.34                                                                  |

## BONUS QUESTION

Assessed on: *undetermined*

Use reactions **A-D** to determine the standard change in enthalpy ( $\Delta H_{\text{rxn}}^{\circ}$ ) for the following reaction.

